

# Groups, rings and modules

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Lecture 1 – 22/01/26

## §1 Groups

### §1.1 Groups – a reminder

**Definition 1.1.** A *group* is a triple  $(G, \cdot, e)$  where  $G$  is a set,

$$\cdot : G \times G \rightarrow G$$

is a function, and  $e \in G$  such that

- *Associativity.*  $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ ;
- *Identity.* For all  $a \in G$ ,  $a \cdot e = e \cdot a = a$ ;
- *Inverses.* For all  $a \in G$ , there exists  $a^{-1} \in G$  such that  $a \cdot a^{-1} = a^{-1} \cdot a = e$ .

The size of  $G$ ,  $|G|$ , is called the *order* of the group.

### Proposition 1.2 (Uniqueness of inverses)

Inverses are unique.

*Proof.* Left as an exercise (see 1A). ■

**Definition 1.3** (Subgroups). If  $G$  is a group and  $H \subseteq G$ , then we say  $H$  is a *subgroup* if

- $e \in H$ ;
- for all  $a, b \in H$ ,  $a \cdot b \in H$ ;
- $(H, \cdot, e)$  is a group.

In order to say item 3, we must first establish 1 and 2.

**Proposition 1.4**

A non-empty subset  $H \subseteq G$  is a subgroup if and only if, for all  $h_1, h_2 \in H$ ,  $h_1 h_2^{-1} \in H$ .

*Proof.* Left as an exercise (see 1A). ■

**Definition 1.5.** A group  $G$  is called *abelian* if, for all  $g_1, g_2 \in G$ ,  $g_1 g_2 = g_2 g_1$ .

**Example 1.6** • We have  $(\mathbb{Z}, +)$ ,  $(\mathbb{Q}, +)$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ ,  $\mathbb{Z}/n\mathbb{Z}$ .

- The symmetric groups  $S_n$ : permutations of  $\{1, \dots, n\}$  under composition. More generally for any set  $X$ , we have  $\text{Sym}(X)$ .
- The dihedral groups  $D_{2n}$ : the group of symmetries of a regular  $n$ -gon.
- The general linear group over  $\mathbb{R}$ ,  $\text{GL}(n, \mathbb{R})$  or  $\text{GL}_n(\mathbb{R})$ , is the set of  $\{f : \mathbb{R}^n \rightarrow \mathbb{R}^n \mid f \text{ is an } \mathbb{R}\text{-linear bijection}\}$ . Since  $\text{GL}(n, \mathbb{R})$  is made of functions, we can think of it a subset of  $\text{Sym}(\mathbb{R}^n)$ . (Equivalently, it's the set of  $n$  by  $n$  invertible matrices with elements in  $\mathbb{R}$ ).

Some examples of subgroups:

- The even permutations  $A_n \leq S_n$  (the alternating group);
- Rotational symmetries of an  $n$ -gon,  $C_n \leq D_{2n}$ ;
- Determinant one matrices  $\text{SL}(n, \mathbb{R}) \leq \text{GL}(n, \mathbb{R})$ .

Other examples include  $C_2 \times C_2$ ,  $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ .

**Definition 1.7.** Let  $G$  be a group,  $g \in G$  and  $H \leq G$  a subgroup. Then the *left coset*

$$gH = \{gh : h \in H\}$$

Basic properties include that  $g \in gH$ , so cosets are non-empty. Also, cosets partition  $G$  – they have no non-trivial overlaps and every  $g$  lies in some coset. Furthermore, the map

$$\begin{aligned} H &\rightarrow gH \\ h &\mapsto gh \end{aligned}$$

defines a bijection.

**Theorem 1.8 (Lagrange)**

If  $G$  is a finite group and  $H \leq G$  then

$$|G| = |H| \cdot |G : H|$$

where  $|G : H|$ , the index of  $H$  in  $G$ , is the number of cosets of  $H$  in  $G$ .

**Definition 1.9.** Let  $G$  be a group and let  $g \in G$ . The *order* of  $g$ ,  $\text{ord}(g)$ , is the smallest positive integer  $n$  such that  $g^n = e$ , or infinity if no such  $n$  exists.

**Proposition 1.10**

Let  $g \in G$ . Then,  $\text{ord}(g) \mid |G|$ .

*Proof.* Let  $g \in G$  and consider  $H = \{e, g, g^2, \dots, g^{n-1}\}$  where  $n = \text{ord}(g)$ . This is a subgroup: it is non-empty, and  $g^r \cdot g^{-s} = g^{r-s} \in H$  (after potentially multiplying by  $g^n = e$ ). Also, the elements of  $H$  are distinct – if  $g^i = g^j$  for  $0 \leq i < j \leq n$ , then  $g^{j-i} = e$ , contradicting the minimality of  $\text{ord}(g)$ . Hence  $|H| = \text{ord}(g)$  – now apply Lagrange. ■

**§1.2 Normal subgroups, quotients and homomorphisms**

Let  $H \leq G$ . When are two cosets of  $H$  in  $G$  equal? If  $gH = g'H$ , then  $g \in g'H$ , so  $g = g'h$  for some  $h \in H$ . In other words,  $gH = g'H \iff (g')^{-1}g \in H$ .

We denote the set of left cosets of  $H \leq G$  by  $G/H = \{gH : g \in G\}$ . Does  $G/H$  carry a natural group structure?

**Lecture 2 – 24/01/26**

We propose the ‘natural’ group structure:

$$g_1H \cdot g_2H = (g_1g_2)H$$

But this rule may depend on the coset representatives we pick. We must check that it doesn't. Suppose  $g_2H = g'_2H$ , so  $g'_2 = g_2h$  for some  $h \in H$ . Then

$$g_1H \cdot g'_2H = (g_1g'_2)H = g_1g_2hH = g_1g_2H$$

On the other hand, if  $g_1H = g'_1H$ , so  $g'_1 = g_1h$  for  $h \in H$ . Then

$$g'_1H \cdot g_2H = g_1hg_2H$$

and that is not automatically equal to  $g_1g_2H$  – we need  $g_2^{-1}hg_2 \in H$  to hold for all  $g_2 \in G, h \in H$ .

**Definition 1.11 (Normal subgroups).** A subgroup  $H \leq G$  is *normal* if, for all  $g \in G, h \in H, ghg^{-1} \in H$ . We write this as  $H \triangleleft G$ .

**Remark 1.12.** When  $H \triangleleft G$ , then  $G/H$  inherits a group structure by the rule above. We will refer to this as the *quotient group of  $G$  by  $H$* .

**Definition 1.13 (Homomorphisms).** Let  $G$  and  $H$  be groups. A homomorphism from  $G$  to  $H$  is a function  $f : G \rightarrow H$  such that, for all  $g_1, g_2 \in G$ ,  $f(g_1g_2) = f(g_1)f(g_2)$ .

Results such as  $f(e_G) = e_H$  follow immediately.

**Lemma 1.14**

If  $f : G \rightarrow H$  is a homomorphism, then  $f(g^{-1}) = f(g)^{-1}$ .

*Proof.* Calculate  $f(g \cdot g^{-1})$  in two ways.

$$\textcircled{1} \quad f(gg^{-1}) = f(e_G) = e_H;$$

$$\textcircled{2} \quad f(gg^{-1}) = f(g)f(g^{-1})$$

so  $f(g)f(g^{-1}) = e_H$  and the result follows. ■

**Definition 1.15.** Let  $f : G \rightarrow H$  be a homomorphism. The *kernel* of  $f$  is

$$\ker f = \{g \in G : f(g) = e_H\}$$

and the *image* of  $f$  is

$$\text{im } f = \{h \in H : h = f(g) \text{ for some } g\}$$

**Proposition 1.16**

Let  $f : G \rightarrow H$  be a homomorphism. Then  $\ker f \triangleleft G$  and  $\text{im } f \leq H$ .

*Proof.* Observe  $e \in \ker f$  so  $\ker f$  non-empty. Now, if  $x, y \in \ker f$ , then

$$f(xy^{-1}) = f(x)f(y)^{-1} = e$$

so  $xy^{-1} \in \ker f$ . For normality, let  $x \in G$  and  $g \in \ker f$ . Now

$$f(xgx^{-1}) = f(x)ef(x)^{-1} = e$$

so  $xgx^{-1} \in \ker f$  and hence  $\ker f \triangleleft G$ .

For the image, observe that, if  $a, b \in \text{im } f$  with  $a = f(x)$ ,  $b = f(y)$ , then  $ab^{-1} = f(xy^{-1}) \in \text{im } f$ . ■

**Remark.** Observe  $f : G \rightarrow G/H$  taking  $g \mapsto gH$  is a homomorphism. This is a nice way of expressing its ‘naturalness’.

**Definition 1.17.** A homomorphism  $f : G \rightarrow H$  is an *isomorphism* if it is a bijection. Two groups  $G$  and  $H$  are said to be *isomorphic*,  $G \cong H$ , if there exists an isomorphism between them.

**Theorem 1.18 (First isomorphism theorem)**

Let  $f : G \rightarrow H$  be a homomorphism, then  $\ker f$  is normal in  $G$  and the homomorphism

$$\begin{aligned} G/\ker f &\rightarrow \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

is an isomorphism. In particular,  $G/\ker f \cong \operatorname{im} f$ .

*Proof.* We have already shown the normality result. Consider the map

$$\begin{aligned} \varphi : G/\ker f &\rightarrow \operatorname{im} f \\ g\ker f &\mapsto f(g) \end{aligned}$$

- $\varphi$  is well-defined. If  $g\ker f = g'\ker f$ , then  $g'g^{-1} \in \ker f$ , so  $f(g'g^{-1}) = e$  which gives  $f(g) = f(g')$  as required.
- $\varphi$  is surjective. Let  $h \in \operatorname{im} f$ . Then  $h = f(g)$  for some  $g \in G$ , and so  $h = \varphi(g\ker f)$ .
- $\varphi$  is injective. Suppose  $\varphi(g\ker f) = \varphi(g'\ker f)$ , then  $f(g) = f(g')$ . But then  $g'g^{-1} \in \ker f$ , so  $g\ker f = g'\ker f$ .

■

**Example**

Consider the homomorphism

$$\begin{aligned} \exp : (\mathbb{C}, +) &\rightarrow (\mathbb{C}^*, \times) \\ z &\mapsto e^z \end{aligned}$$

Then  $\ker \exp = 2\pi i\mathbb{Z}$ , so  $\mathbb{C}^* \cong \mathbb{C}/2\pi i\mathbb{Z}$ .

**Theorem 1.19 (Second isomorphism theorem)**

Let  $H \leq K$  and  $K \triangleleft G$ . Then

$$HK = \{hk : h \in H, k \in K\}$$

is a subgroup of  $G$ . Furthermore,  $H \cap K$  is normal in  $H$ , and there is an isomorphism  $HK/K \cong H/H \cap K$ .

*Proof.* We will show the results in order.

- $HK$  is a subgroup. It contains  $e_G$ , so is non-empty. Now consider  $x = hk$  and  $y = h'k'$ . We'll show  $yx^{-1} \in HK$ . Observe

$$\begin{aligned} yx^{-1} &= h'k'(hk)^{-1} \\ &= h'k'k^{-1}h^{-1} \end{aligned}$$

$$\begin{aligned} &= (h'h^{-1})h(k'k^{-1})h^{-1} \\ &= h'h^{-1}k'' \end{aligned}$$

by normality of  $K$ .

- $H \cap K$  is a kernel. Consider

$$\begin{aligned} \varphi : H &\rightarrow G/K \\ h &\mapsto hK \end{aligned}$$

by restricting the domain from  $G \rightarrow G/K$  – the kernel is  $H \cap K$ .

- *Final statement.* By the first isomorphism theorem, we have

$$\text{im } \varphi \cong H/(H \cap K)$$

so we must show  $\text{im } \varphi \cong HK/K$ . But  $\text{im } \varphi$  is the set of  $K$ -cosets in  $G$  that are representable by elements of  $H$ , i.e.  $\{hK : h \in H\} \subseteq G/K = HK/K$ . ■

### §1.2.1 Correspondence between subgroups of $G$ and $G/K$

Let  $p : G \rightarrow G/K$  be the quotient homomorphism. We have a bijection

$$\begin{aligned} \{\text{subgroups of } G \text{ containing } K\} &\leftrightarrow \{\text{subgroups of } G/K\} \\ K \triangleleft L \leq G &\mapsto L/K \leq G/K \\ \{g \in G : gK \in A\} &\leftrightarrow A \leq G/K \end{aligned}$$

The forward bijection is just applying the projection map on a subset, and, since the quotient map is a homomorphism, its image is a subgroup; and the backward bijection is actually just applying the preimage of  $p$ . It is an exercise to check that the forward and backward directions are inverses.

**Remark.** In fact the correspondence matches normal subgroups on the two sides.

#### **Theorem 1.20 (Third isomorphism theorem)**

Let  $K \leq L \leq G$  be normal subgroups of  $G$ . Then there exists an isomorphism  $(G/K)/(L/K) \cong G/L$ .

*Proof.* Define a map

$$\begin{aligned} \varphi : G/K &\rightarrow G/L \\ gK &\mapsto gL \end{aligned}$$

- $\varphi$  is well-defined. If  $gK = g'K$ , then  $g'g^{-1} \in K \subseteq L \implies gL = g'L$ .
- $\varphi$  is a homomorphism. Clear.

We have  $\ker \varphi = \{gK : gL = L\} = \{gK : g \in L\} = L/K$ . Hence we are done by the first isomorphism theorem. ■

**Definition 1.21 (Simplicity).** A group  $G$  is called *simple* if its only normal subgroups are  $G$  and  $\{e\}$ .

**Proposition 1.22**

Let  $G$  be an abelian group. Then  $G$  is simple if and only if  $G \cong C_p$  for some prime  $p$ .

*Proof.* If  $G \cong C_p$ , then any non-identity element generates  $G$  by, for example, Lagrange. So  $G$  is simple.

If  $G$  is simple and abelian, then take any  $g \neq e \in G$  and consider  $\langle g \rangle$ . This is a subgroup, and since  $G$  is abelian, it is automatically normal. Hence, by simplicity,  $\langle g \rangle = G$ , and so  $G$  is cyclic.

If  $G$  is infinite and cyclic, then  $G \cong \mathbb{Z}$ , but  $\mathbb{Z}$  is not simple, so this is a contradiction. Hence  $G$  is finite and  $G \cong C_m$ . If  $q$  divides  $m$ , take the subgroup generated by  $g^{m/q}$ , where  $g$  generates  $G$ . Normality is automatic, so  $q = 1$  or  $q = m$  by simplicity – in other words,  $m$  is prime. ■

**Theorem 1.23**

Let  $G$  be a finite group. Then, there exist subgroups

$$G = H_1 \triangleright H_2 \triangleright H_3 \triangleright \cdots \triangleright H_n = \{e\}$$

such that  $H_i/H_{i+1}$  is simple for all  $i$ .

*Proof.* If  $G$  is simple,  $H_2 = \{e\}$ , and we're done. Otherwise, let  $H_2$  be a proper normal subgroup of maximal order in  $G$ . We claim  $G/H_2$  is simple. If not, consider the quotient map  $\pi : G \rightarrow G/H_2$ .

If  $N \triangleleft G/H_2$  is a proper normal subgroup, by the correspondence theorem, we can find a proper normal subgroup in  $G$  containing  $H_2$ . But this contradicts the maximality of  $H_2$ .

Now we can just iterate – note that  $|H_{i+1}| < |H_i|$ , so we will eventually terminate. ■

**Remark.** This sequence is not necessarily unique.

**§1.3 Group actions and permutations**

**Definition 1.24.** Let  $X$  be a set. Let  $\text{Sym}(X)$  be the set of permutations on  $X$ . The identity is the identity function, and the group law is given by composition. If  $X = \{1, \dots, n\}$ , then we write  $\text{Sym}(X) = S_n$ .

- It's convenient to write  $\sigma \in S_n$  as a product of disjoint cycles.
- Every  $\sigma \in S_n$  is a product of transpositions, and the number of such transpositions gives rise to the *sign* of  $\sigma$ .

$$\text{sign} : S_n \rightarrow \{\pm 1\}$$

$$\sigma \mapsto \begin{cases} +1 & \sigma \text{ is a product of an even number of transpositions} \\ -1 & \sigma \dots \text{ odd} \dots \end{cases}$$

sign is a homomorphism, and is surjective when  $n \geq 3$ .

Let  $X$  be a set and  $G$  a group, and consider a homomorphism  $G \rightarrow \text{Sym}(X)$ . This is called a *permutation representation* of  $G$ .

**Definition 1.25** (Group actions). An *action* of a group  $G$  on a set  $X$  is a function

$$\begin{aligned} G \times X &\rightarrow X \\ (g, x) &\mapsto g \cdot x \end{aligned}$$

such that

- $e \cdot x = x$  for all  $x \in X$ ;
- $g_1 \cdot (g_2 x) = (g_1 g_2) \cdot x$  for all  $g_1, g_2 \in G$  and  $x \in X$ .

We say  $G \curvearrowright X$ .

We should note the bijection between permutation representations and group actions.

**Proposition 1.26**

The map  $a$  is a bijection.

$$\begin{aligned} a : \{\text{homomorphisms } G \rightarrow \text{Sym}(X)\} &\longrightarrow \{\text{actions } G \curvearrowright X\} \\ \varphi : G \rightarrow \text{Sym}(X) &\longmapsto a(\varphi) : G \times X \rightarrow X : (g, x) \mapsto \varphi(g) \cdot x \end{aligned}$$

*Proof.* Let us simply construct an inverse. Let  $\cdot : G \times X \rightarrow X$  be an action. Define

$$\begin{aligned} \varphi : G &\rightarrow \text{Sym}(X) \\ g &\mapsto \varphi(g) := [g \cdot : X \rightarrow X] \end{aligned}$$

This is a permutation as  $(g^{-1}) \cdot : X \rightarrow X$  is the inverse. Indeed,

$$\begin{aligned} \varphi(g^{-1})(\varphi(g)(x)) &= g^{-1} \cdot (g \cdot x) \\ &= (g^{-1}g)x \\ &= x \end{aligned}$$

To see that this rule gives a homomorphism is simple. ■

We have some notation:

- $G^X = \text{im } \phi \leq \text{Sym}(X)$ ;
- $G_X = \ker \phi \triangleleft G$

It is a special case of the first isomorphism theorem that  $G^X \cong G/G_X$ .

**Example 1.27**

Let  $G$  be the symmetry group of the unit cube, and let  $X$  be the set of the cube's diagonals. Clearly  $G$  acts on  $X$ . We have hence

$$\phi : G \rightarrow \text{Sym}(X) \cong S_4$$

We can see  $\text{im } \phi \cong S_4$ , since we can swap any two diagonals – then these transpositions generate the whole symmetric group. Furthermore,  $\ker \phi = \{e, s\}$ , where  $s$  sends every vertex to its opposite. As an immediate consequence, we conclude  $|G| = 48$ .



**Example 1.28**

For any group  $G$ , we can make  $G$  act on itself (as a set) by

$$\begin{aligned} G \times G &\rightarrow G \\ (g, g_1) &\mapsto gg_1 \end{aligned}$$

which gives  $\phi : G \rightarrow \text{Sym}(G)$ . The kernel  $\ker \phi = \{e\}$ , and so  $\phi$  is injective – we conclude that  $G$  is isomorphic to a subgroup of  $\text{Sym}(G)$  by the first isomorphism theorem.

**Example 1.29**

Let  $H \leq G$  and  $X = G/H$ . Then  $G \curvearrowright X$  by

$$(g, g_1H) \rightarrow gg_1H$$

We get a homomorphism  $\phi : G \rightarrow \text{Sym}(X)$ . Consider the kernel – it is clear that  $\ker \phi \subseteq H$ . But more is true. If  $g \in \ker \phi$ , then

$$\begin{aligned} gg_1H &= g_1H \\ \implies g_1^{-1}gg_1 &\in H \\ \implies \ker \phi &\subseteq \bigcap_{g_1 \in G} g_1Hg_1^{-1} \end{aligned}$$

that is,  $g$  is contained in every conjugate of  $H$ . Furthermore, this containment is reversible: suppose  $g \in \bigcap_{g_1 \in G} g_1Hg_1^{-1}$ . Then  $gg_1H = g_1H$ , so  $g \in \ker \phi$ . We conclude

$$\ker \phi = \bigcap_{g_1 \in G} g_1Hg_1^{-1}$$

If  $H$  were normal, we would just have  $H$  on the right-hand side – but when  $H$  is not normal, we have an interesting way to make a smaller normal subgroup,  $\ker \phi$ , out of  $H$ .

**Theorem 1.30**

Let  $G$  be finite and let  $H \leq G$  be of index  $n$ . Then, there exists a normal  $K \triangleleft G$  with  $K \subseteq H$  such that  $G/K$  is isomorphic to a subgroup of  $S_n$ . Thus  $|G/K|$  divides  $n!$ , and  $|G/K| \geq n$ .

*Proof.* Consider  $\phi : G \rightarrow \text{Sym}(G/H)$  from the example. The kernel, call it  $K$ , is normal. We have shown it is contained in  $H$  already – applying the first isomorphism theorem, we have

$$G/K \cong \text{im } \phi \leq \text{Sym}(G/H) \cong S_n$$

Lagrange gives  $|G/K|$  divides  $n!$ , and  $K \leq H$ , so  $|G/K| \geq |G/H| = n$ . ■

**Corollary 1.31**

Let  $G$  be a non-abelian simple group. Let  $H \leq G$  be a proper subgroup with index  $n > 1$ . Then  $G$  is isomorphic to a subgroup of  $A_n$  with  $n \geq 5$ .

**Lecture  $n$** **Lecture  $n + 1$** 

*Claim 3.* Any non-trivial  $H \triangleleft A_n$  contains a 3-cycle.

- If  $H$  contains an element of the form  $\sigma = (1\ 2\ \dots\ \gamma)\tau$  with  $\tau$  disjoint from  $(1\ 2\ \dots\ \gamma)$  and  $\gamma \geq 4$ , then normality of  $H$  gives that  $\delta^{-1}\sigma\delta \in H$ , so furthermore  $\sigma^{-1}\delta^{-1}\sigma\delta \in H$ . Since  $\tau$  misses 1, 2, 3, it commutes with  $\delta$ , and with  $(1\ 2\ \dots\ \gamma)$  by assumption.

Then, expanding  $\sigma^{-1}\delta^{-1}\sigma\delta$  out, we get  $(2\ 3\ \gamma)$ , so we have our 3-cycle.

- If  $H$  contains an element of the form  $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$  in a disjoint cycle decomposition, take  $\delta = (1\ 2\ 4)$  and calculate  $\sigma^{-1}\delta^{-1}\sigma\delta$  – this is a 5-cycle, bringing us into case 1.
- If  $H$  contains  $\sigma = (1\ 2\ 3)\tau$  where  $\tau$  is a product of disjoint transpositions (if  $\tau$  contained anything longer, we could relabel and apply cases 1 or 2). Then  $\sigma^2$  is a 3-cycle.
- If  $H$  contains  $\sigma = (1\ 2)(3\ 4)\tau$  where  $\tau$  is a product of disjoint transpositions, let  $\delta = (1\ 2\ 3)$  and consider  $\mu = \sigma^{-1}\delta^{-1}\sigma\delta = (1\ 4)(2\ 3)$ . Then let  $\nu = (1\ 5\ 2)\mu(1\ 2\ 5) = (1\ 3)(4\ 5)$  – since  $\mu\nu = (1\ 2\ 3\ 4\ 5)$ , we are in case 1.

This covers all cases up to relabelling.

**§1.4 Finite  $p$ -groups**

Let  $p$  be a prime. Groups of order  $p^n$  for some  $n$  are called  $p$ -groups.

**Theorem 1.32**

Let  $G$  be a  $p$ -group. Then the centre of  $G$  is non-trivial.

*Proof.* Let  $G$  act on itself by conjugation. The centre  $Z(G)$  is the set of all size one orbits. Since the orbits partition  $G$ , we have

$$|G| = p^n = |Z(G)| + \sum \text{sizes of conjugacy classes of size } > 1$$

We know conjugacy classes have sizes dividing  $p^n$ . Hence, if a conjugacy class has size larger than 1, it is a multiple of  $p$ . Then  $Z(G)$  is a difference of multiples of  $p$  and the result follows ■

**Theorem 1.33**

A group of size  $p^2$  is abelian.

**Lemma 1.34**

If  $G/Z(G)$  is cyclic, then  $G$  is abelian.

*Proof of lemma.* We have a coset  $xZ(G)$  in  $G/Z(G)$  that generates  $G/Z(G)$ , so every coset has the form  $x^mZ(G)$ . This means every  $G$  can be written as  $g = x^mz$  for some  $m$  and  $z \in Z(G)$ . From this the result immediately follows. ■

*Proof of theorem.* By [theorem 1.32](#),  $Z(G)$  is either  $G$  or has size  $p$ . In the former case we are done – in the latter case,  $G/Z(G) \cong C_p$ , which is cyclic, so, by [lemma 1.34](#),  $G$  is abelian. ■

**Theorem 1.35**

Let  $G$  be a  $p$ -group of size  $p^n$ . Then  $G$  has a subgroup of size  $k$  for all  $0 \leq k \leq n$ .

*Proof.* We induct on  $n$ . *Base case.*  $n = 1$ , the result is clear. *Inductive step.* Suppose  $n > 1$ . If  $k = 0$ , take  $\{e\}$ . Otherwise, note  $Z(G)$  is non-trivial. Let  $x \in Z(G)$  with  $x \neq e$ , and its order is a power of  $p$  – if we replace  $x$  with some power of  $x$ , say  $x'$ , we can produce an element of order  $p$ . Then  $\langle x' \rangle'$  has order  $p$  and is normal.

Now consider  $G/\langle x' \rangle'$ . This is a group of size  $p^{n-1}$ , and so it contains a subgroup, say  $L$ , of size  $p^{k-1}$  by induction. By the correspondence of subgroups, we find  $K \leq G$  containing  $\langle x' \rangle'$  such that  $K/\langle x' \rangle' \cong L$ . But then  $|K| = p^k$ , as required. ■

**§1.5 Finite abelian groups****Theorem 1.36 (Classification of finite abelian groups)**

Let  $G$  be a finite abelian group. There exist positive integers  $d_1, \dots, d_r$  such that

$$G \cong C_{d_1} \times \cdots \times C_{d_r}$$

Furthermore, we can choose the  $d_i$  such that  $d_{i+1} \mid d_i$ , in which case the expression is unique.

**Example 1.37**

Consider, for example, abelian groups of order 8 – they are exactly  $C_8$ ,  $C_4 \times C_2$ , and  $C_2 \times C_2 \times C_2$ .

**Lemma 1.38** (Chinese remainder theorem)

If  $n$  and  $m$  are coprime, then  $C_n \times C_m \cong C_{nm}$ .

*Proof.* It suffices to find an order  $mn$  element of  $C_n \times C_m$ . If  $g \in C_n$  has order  $n$  and  $h \in C_m$  has order  $m$ , then consider  $(g, h) \in C_n \times C_m$ . Suppose it has order  $k$ , then  $g^k = h^k = e$ , so that  $n \mid k$  and  $m \mid k$ . But then  $nm \mid k$ , and  $k \mid nm$  by Lagrange, so  $k = mn$  as required. ■

**Lecture 7 – 05/02/26****Corollary 1.39**

Let  $G$  be finite and abelian, then we have

$$G \cong C_{l_1} \times \cdots \times C_{l_n}$$

where each  $l_i$  is a prime power.

*Proof.* The result follows from [theorem 1.36](#) and [lemma 1.38](#). ■

**§1.6 Sylow theorems**

Let  $G$  be a finite group of size  $p^a m$  where  $p \nmid m$  and  $p$  is some prime. A *Sylow subgroup* of  $G$  is a subgroup of order  $p^a$ .

**Theorem 1.40** (Sylow theorems)

Let  $G$  be a finite group and  $p$  a prime.

1. The set  $\text{Syl}_p(G) = \{P \leq G : P \text{ is Sylow-}p\}$  is non-empty;
2. any  $H, H' \in \text{Syl}_p(G)$  are conjugate;
3. if  $n_p = |\text{Syl}_p(G)|$ , then  $n_p \equiv 1 \pmod{p}$ , and  $n_p \mid |G|$  (so  $n_p \mid m$ ).

**Corollary 1.41**

$\text{Syl}_p(G) = \{P\}$  if and only if  $P$  is normal.

*Proof.* For any  $g \in G$ ,  $gPg^{-1}$  is isomorphic as a group to  $P$ , so it is also a Sylow  $p$ -subgroup. ■

### Corollary 1.42

Let  $G$  be non-abelian and simple and suppose  $p \mid |G|$ . Then  $|G|$  divides  $\frac{n_p!}{2}$  and  $n_p \geq 5$ .

*Proof.* The conjugation action of  $G$  on  $\text{Syl}_p(G)$  gives a homomorphism

$$\phi : G \rightarrow \text{Sym}(\text{Syl}_p(G)) \cong S_{n_p}$$

By simplicity of  $G$ ,  $\ker \phi = \{e\}$  or  $G$ . If  $\ker \phi = G$ , then  $gPg^{-1} = P$  for all  $g \in G$ , so  $P$  is normal. Since  $P \neq \{e\}$ ,  $P = G$ . But  $Z(P) \triangleleft P$  non-trivially, which is a contradiction. Hence  $\phi$  is injective, so  $G$  is isomorphic to a subgroup of  $S_{n_p}$ . Consider now

$$G \xrightarrow{\phi} S_{n_p} \xrightarrow{\text{sign}} \{\pm 1\}$$

If the composition is surjective, then  $\ker(\text{sign} \circ \phi)$  is an index 2 subgroup of  $G$ . But  $G$  is simple, so this is impossible, (since  $G \neq C_2$ ). In other words,  $\text{im } \phi \subseteq A_{n_p}$ , and the result follows by Lagrange. Finally, all non-abelian subgroups of  $A_2$ ,  $A_3$  and  $A_4$  are not simple. ■

### Example

Suppose  $G$  has size  $11 \times 12 = 132$ . If  $G$  is simple, then none of the  $n_p$ 's are equal to 1. Furthermore, there are 12 Sylow-11 subgroups, since  $n_1 \equiv 1 \pmod{11}$  and  $n_1 \mid 12$ .

Similarly,  $n_3 \equiv 1 \pmod{3}$ , and  $n_3 \mid 44$ , so  $n_3 = 4$  or  $22$  – by [corollary 1.42](#),  $n_3 = 22$ . Any two Sylow 11-subgroups intersect only at  $\{e\}$ , so we already have 120 elements. We also have at least 44 elements of order 3, which is impossible, so no such simple  $G$  exists.

*Proof of Sylow part 1.* We will try and produce a Sylow  $p$ -subgroup as a stabiliser of an element in an appropriate action.

Let  $|G| = n = p^a m$ , of course  $p \nmid m$ . Let  $\Omega = \{X \subseteq G : |X| = p^a\}$ . Let  $G$  act on  $\Omega$  by  $g \cdot \{g_1, \dots, g_{p^a}\} = \{gg_1, \dots, gg_{p^a}\}$ .

*Claim 1.* We have  $|\Omega| = \binom{n}{p^a} = m \not\equiv 0 \pmod{p}$ .

*Proof.* Expand  $(1+x)^n \pmod{p}$ .

*Claim 2.* Suppose  $U \in \Omega$  and  $H \leq G$  stabilises  $U$ , then  $|H| \mid |U|$ .

*Proof.* We have  $hU = U$  for all  $h \in H$ . In other words, for all  $u \in U$ , the coset  $Hu$  is contained in  $U$ . Furthermore, every  $u \in U$  lies in some coset of  $H$ , and the cosets are disjoint, so the cosets partition  $U$ . We get  $|H| \mid |U|$ .

We now know  $|\Omega| \not\equiv 0 \pmod{p}$ , and  $|\Omega| = |O_1| + \dots + |O_r|$  where the  $O_i$  are the orbits of the action. Since  $p \nmid |\Omega|$ , there exists an orbit, say  $\Theta$ , whose size is coprime to  $p$ .

Let  $T \in \Theta$ . By the orbit-stabiliser theorem,  $|G| = |\Theta| \cdot |\text{Stab}(T)|$ , so  $p^a m = |\Theta| |\text{Stab}(T)|$ . By claim 2,  $|\text{Stab}(T)|$  divides  $p^a$ , but there are no factors of  $p$  in  $\Theta$ , so we are done. ■

## §2 Rings

### Lecture $n + 1$

#### §2.1 Basic definitions

Let  $R$  be a ring. We have

$$\begin{aligned} 0 + 0 &= 0 \\ \implies r \cdot 0 + r \cdot 0 &= r \cdot 0 \\ \implies r \cdot 0 &= 0 \end{aligned}$$

Furthermore, if  $1 = 0$ , we have  $r = 1 \cdot r = 0 \cdot r = 0$ , so we have the *zero ring*.

**Definition 2.1** (Ring direct product). Let  $R, S$  be rings. The *direct product*  $R \times S$  is a ring with operations and identities defined pointwise.

**Definition 2.2** (Polynomials). Let  $R$  be a ring. A *polynomial* in  $X$  with coefficients in  $R$  is an expression of the form

$$f(X) = a_0 + a_1X + \cdots + a_nX^n$$

where each  $a_i \in R$ , and the  $X^i$  are formal symbols. As convention, we will identify  $f(X)$  with  $f(X) + 0 \cdot X^{n+1}$  – then, the largest  $n$  with  $a_n \neq 0$  is the *degree* of  $f(x)$  (by convention, we say the degree of 0 is  $-\infty$ ). Finally, a polynomial is *monic* if it has degree  $n$  and  $a_n = 1$ .

**Definition 2.3** (Polynomial rings). Let  $R$  be a ring. The *polynomial ring*  $R[X]$  is the set of all polynomials with coefficients in  $R$ , with obvious operations and identities.

**Remark.** Note that, for every  $f \in R[X]$ , we get a function  $R \rightarrow R$  which takes  $r$  to  $f(r)$ . But this is not the same as the polynomial itself.

**Definition 2.4** (Rings of formal power series). The ring of *formal power series* over  $R$  in the variable  $X$  is

$$R[[X]] = \left\{ \sum_{i=0}^{\infty} r_i X^i : r_i \in R \right\}$$

with obvious operations and identities.

**Example 2.5**

Consider the element  $1 - x \in R[X]$ . Is it a unit?

Suppose we have  $(1 - x)g(x) = 1$ , where  $g(x) = a_0 + \cdots + a_n x^n$ , with  $a_n \neq 0$ . Then

$$(1 - x)g(x) = a_0 + (a_1 - a_0)x + \cdots + (a_n - a_{n-1})x^n - a_n x^{n+1}$$

and then, by comparing the  $x^{n+1}$  coefficient, we can see this is not 1.

On the other hand, if we view  $1 - x \in R[[X]]$ , this has an inverse:

$$(1 - x)(1 + x + x^2 + \cdots) = 1$$

In fact, every power series has an inverse as long as  $r_0$  is invertible.

**Definition 2.6** (Rings of Laurent series). Write  $R[X, X^{-1}]$  for the ring of *Laurent series* with coefficients in  $R$ , that is

$$f = \sum_{i \in \mathbb{Z}} a_i x^i$$

with  $a_i \in R$  and all but finitely many  $a_i = 0$ .

**§2.2 Homomorphisms, ideals and quotients**

**Definition 2.7** (Ring homomorphisms). Let  $R$  and  $S$  be rings. A function  $f : R \rightarrow S$  is a *ring homomorphism* if

- $f(r_1 + r_2) = f(r_1) + f(r_2)$ ;
- $\implies f(0) = 0$ ;
- $f(r_1 r_2) = f(r_1) f(r_2)$ ;
- $\nRightarrow f(1) = 1$ .

The kernel of  $f$  is  $\ker f = \{r \in R : f(r) = 0\}$ , and the image of  $f$  is  $\operatorname{im} f = \{s \in S : s = f(r) \text{ for some } r \in R\}$ .

**Definition 2.8** (Ring isomorphisms). If  $f : R \rightarrow S$  is a bijective ring homomorphism, then it is called an isomorphism. Furthermore, its inverse is also a ring homomorphism.

**Definition 2.9.** A subset  $I \subseteq R$  is called an *ideal*, written  $I \triangleleft R$ , if

- $I$  is a subgroup additively;
- if  $a \in I$  and  $b \in R$ , then  $ab \in I$ .

Usually ideals are not subrings, since they can only contain 1 if  $I = R$ .

**Lemma 2.10**

If  $\phi : R \rightarrow S$  is a homomorphism, then  $\ker \phi \triangleleft R$ .

*Proof.* Since  $\phi$  is a group homomorphism, we know  $\ker \phi$  is a subgroup.

Furthermore, if  $a \in \ker \phi$  and  $b \in R$ , then  $\phi(ab) = \phi(a)\phi(b) = 0 \cdot \phi(b) = 0$ . ■

**Remark.** If  $a \in R$  is a unit, and  $I \triangleleft R$  is an ideal, then  $u \in I \implies I = R$  – since  $1 = uv$ ,  $1 \in I$ , and so  $I = R$ .

### Example 2.11

Observe that  $n\mathbb{Z}$  is an ideal of  $\mathbb{Z}$  for any  $n$ . In fact, every ideal of  $\mathbb{Z}$  has this form. Indeed, if  $I \neq \{0\}$ , let  $n \in \mathbb{Z}$  be the minimal positive element of  $I$ . By the Euclidean algorithm, we can write  $m \in I$  as  $m = qn + r$ , with  $0 \leq r < n$ , and  $qn \in I$ , so  $r \in I$  – this contradicts minimality of  $n$ .

**Definition 2.12.** Let  $A \subseteq R$  be a subset. The *ideal generated by  $A$*  is

$$\langle A \rangle = \left\{ \sum_{a \in A} r_a a : r_a \in R, r_a = 0 \text{ for all but finitely many } a \right\}$$

**Definition 2.13.** An ideal is called *principal* if there exists  $r \in R$  such that  $I = \langle r \rangle$ .

### Example 2.14

Consider the subset of  $R[x]$  given by

$$\{f(x) \in R[x] : f(0) = 0\}$$

which is generated by  $x \in R[x]$  – hence, this is a principal ideal.

**Definition 2.15 (Quotients).** Let  $I \triangleleft R$  be an ideal. The *quotient ring*  $R/I$  is the set of cosets of  $I$  in  $R$  with 0 and 1 given by  $0_R + I$  and  $1_R + I$ , with the rules

$$\begin{aligned} (r_1 + I) + (r_2 + I) &= (r_1 + r_2) + I \\ (r_1 + I) \cdot (r_2 + I) &= r_1 r_2 + I \end{aligned}$$

### Proposition 2.16

The quotient ring is a ring. Furthermore, the function

$$\begin{aligned} f : R &\rightarrow R/I \\ r &\mapsto r + I \end{aligned}$$

is a ring homomorphism.

*Proof.* This is obviously a group homomorphism. Furthermore, multiplication is well-defined: suppose  $r_1 + I = r'_1 + I$ ,  $r_2 + I = r'_2 + I$ . We then have  $r_i - r'_i \in I$ , say  $r_i - r'_i = a_i$ . We then have

$$r'_1 r'_2 = (r_1 + a_1)(r_2 + a_2) = r_1 r_2 + \underbrace{r_1 a_2 + r_2 a_1 + a_1 a_2}_{\in I}$$

We conclude that the coset in  $I$  of  $r'_1 r'_2$  is the same as that of  $r_1 r_2$ .

To check the other ring properties is straightforward. ■



- Example 2.17** • We have  $n\mathbb{Z} \triangleleft \mathbb{Z}$ . The quotient ring  $\mathbb{Z}/n\mathbb{Z}$  consists of cosets of  $n\mathbb{Z}$ , i.e.  $0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \dots, (n-1) + n\mathbb{Z}$ . The ring structure is modular arithmetic.
- Take  $\langle x \rangle \triangleleft \mathbb{C}[x]$ . The elements of the quotient  $\mathbb{C}[x]/\langle x \rangle$  can be represented by elements of the form  $(a_0 + a_1x + \dots + a_nx^n) + \langle x \rangle$  – but all the terms  $a_1x, \dots, a_nx^n$  lie in  $\langle x \rangle$ , so every element is uniquely represented by  $a_0$ . Hence  $\mathbb{C}[x]/\langle x \rangle \cong \mathbb{C}$ .

**Proposition 2.18** (Euclidean algorithm for polynomial rings over fields)

Let  $K$  be a field, and take  $f, g \in K[x]$ . There exist polynomials  $r, q \in K[x]$  such that

$$f = gq + r$$

and  $\deg r < \deg g$ .

*Proof.* Let  $n = \deg f$  and  $m = \deg g$ , so

$$f = \sum_{i=0}^n a_i x^i \quad g = \sum_{i=0}^m b_i x^i$$

with  $a_i, b_i \in K$ ,  $a_n \neq 0$ . If  $n < m$ , set  $q = 0$  and  $r = f$ . On the other hand, if  $n \geq m$ , we proceed by induction on  $n$ . In particular, let  $f_1 = f - a_n b_m^{-1} x^{n-m}$  (note we are using the fact that  $K$  is a field). Now,  $\deg f_1 < n$ . If  $n = m$ , we can write

$$f = (a_n b_m^{-1} x^{n-m})g + f_1$$

Otherwise, if  $n > m$  we can write by induction

$$f_1 = gq_1 + r_1$$

with  $\deg r_1 < m$ . Then

$$f = (a_n b_m^{-1} x^{n-m} + q)g + r_1$$

■

**Corollary**

All ideals in  $K[x]$  for  $K$  a field are principal.

*Proof.* We have proven that Euclidean algorithm  $\implies$  all ideals are principal, since we did it for  $\mathbb{Z}$ . The result follows. ■

**Theorem 2.19** (First isomorphism theorem for rings)

Let  $\phi : R \rightarrow S$  be a ring homomorphism. Then, the map

$$\begin{aligned} f : R/\ker \phi &\rightarrow \text{im } \phi \\ r + \ker \phi &\mapsto \phi(r) \end{aligned}$$

is a well-defined ring isomorphism.

*Proof.* Well-definedness, bijectivity and additivity all follow from the group isomorphism theorem. It remains to check multiplicativity. Observe

$$\begin{aligned} f((r + \ker \phi)(t + \ker \phi)) &= f(rt + \ker \phi) \\ &= \phi(rt) \\ &= \phi(r)\phi(t) \\ &= f(r + \ker \phi)f(t + \ker \phi) \end{aligned}$$

■

**Example 2.20**

Consider the homomorphism

$$\begin{aligned} \phi : \mathbb{R}[x] &\rightarrow \mathbb{C} \\ f(x) &\mapsto f(i) \end{aligned}$$

The map is clearly surjective, and the kernel is exactly those polynomials  $f$  satisfying  $f(i) = 0$ . However, since coefficients of  $f$  are real,  $-i$  is also a root. In conclusion,  $x^2 + 1 \mid f$ , and we conclude  $\ker \phi = \langle x^2 + 1 \rangle$ . Hence, by the first isomorphism theorem,  $\mathbb{C} \cong \mathbb{R}[x]/\langle x^2 + 1 \rangle$ .

**Proposition 2.21** (Second isomorphism theorem for rings)

Let  $R \leq S$  and  $J \triangleleft S$ . Then  $J \cap R$  is an ideal in  $R$ , and

$$(R + J)/J = \{r + J : r \in R\} \leq S/J$$

is a subring. Furthermore,

$$R/(R \cap J) \cong (R + J)/J$$

*Proof.* Define a function

$$\begin{aligned} \phi : R &\rightarrow S/J \\ r &\mapsto r + J \end{aligned}$$

so  $\ker \phi = \{r \in R : r + J = 0\} = R \cap J$ , so this is an ideal by the first isomorphism theorem. Also, the image is  $\text{im } \phi = \{r + J : r \in R\} = (R + J)/J$ . The result follows from the first isomorphism theorem. ■

**Lecture 11 – 14/02/26****Lecture 12 – 17/02/26****Proposition 2.22**

Let  $R$  be a ring. Then  $R$  is a field if and only if the only ideals of  $R$  are  $\{0\}$  and  $R$ .

*Proof.*  $\implies$  If  $R$  is a field and  $I \triangleleft R$ , say  $I \neq \{0\}$ , then  $I$  contains a unit, say  $u \in I$ . But then  $1 \in I$  since  $1 = uv$ . We conclude  $I = R$ .

$\impliedby$  If the only ideals are  $\{0\}$  and  $R$ , take  $0 \neq r \in R$ . Then  $\langle r \rangle$  is  $R$ . So  $1 \in R \implies 1 = rs$ . So  $r$  is a unit and hence  $R$  is a field. ■

**Definition 2.23 (Maximal ideals).** An ideal  $R \neq I \triangleleft R$  is called *maximal* if, for any  $J \triangleleft R$ ,  $I \subseteq J \subseteq R$  implies  $I = J$  or  $J = R$ .

**Proposition 2.24**

An ideal  $I \triangleleft R$  is maximal if and only if  $R/I$  is a field.

*Proof.*  $R/I$  is a field if and only if its only ideals are  $\{0\}$  and  $R/I$  itself. Now apply the ideal correspondence result. ■

**Definition 2.25 (Prime ideals).** An ideal  $R \neq I \triangleleft R$  is *prime* if, whenever  $ab \in I$ , either  $a$  or  $b \in I$ .

**Example 2.26**

An ideal  $n\mathbb{Z} \triangleleft \mathbb{Z}$  is a prime ideal if and only if  $n$  is 0 or a prime.

**Proposition 2.27**

An ideal  $I \triangleleft R$  is prime if and only if  $R/I$  is an integral domain.

*Proof.*  $\implies$  Suppose  $I$  is prime and let  $a + I$  and  $b + I \in R/I$  be cosets. Suppose  $(a + I)(b + I) = ab + I = 0 + I$  if and only if  $ab \in I$ . But, since  $I$  is prime, either  $a$  or  $b \in I$ , so either  $a + I$  or  $b + I = 0 + I$ .

$\impliedby$  If  $R/I$  is an integral domain, then  $ab \in I$  satisfies  $ab + I = (a + I)(b + I) = 0 + I$  in  $R/I$ , so either  $a \in I$  or  $b \in I$ , so  $I$  is prime. ■

**Corollary 2.28**

Every maximal ideal in a ring is prime.

*Proof.* Since every field is an integral domain, the result follows from [proposition 2.24](#) and [proposition 2.27](#). ■

**Remark.** The converse clearly does not hold.

**Proposition 2.29**

The characteristic of an integral domain is either zero or a prime number.

*Proof.* Consider the unique homomorphism  $\mathbb{Z} \rightarrow R$ . Letting the characteristic be  $n$ , we have that  $\langle n \rangle \triangleleft \mathbb{Z}$  is the kernel, so  $\mathbb{Z}/n\mathbb{Z} \leq R$  by the first isomorphism theorem. But then  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain, and the result follows. ■

**§2.3 Factorisation in integral domains**

Throughout this section,  $R$  is an integral domain.

**Definition 2.30 (Division).** Let  $a, b \in R$ . We say  $a$  *divides*  $b$ , written  $a \mid b$ , if there exists  $c \in R$  such that  $ac = b$ . Equivalently, we can say  $\langle b \rangle \subseteq \langle a \rangle$ .

**Definition 2.31 (Associates).** We say  $a, b \in R$  are *associates* if  $a = bc$  for some unit  $c \in R$ . Equivalently, we can say  $a \mid b$  and  $b \mid a$ , or  $\langle a \rangle = \langle b \rangle$ .

**Definition 2.32 (Irreducible elements).** An element  $r \in R$  is *irreducible* if  $r \neq 0$ ,  $r$  is not a unit, and, if  $r = xy$ , either  $x$  or  $y$  is a unit.

**Definition 2.33 (Prime elements).** An element  $r \in R$  is *prime* if  $r \neq 0$ ,  $r$  is not a unit, and, if  $r \mid xy$ , either  $r \mid x$  or  $r \mid y$ .

**Proposition 2.34**

Let  $r \in R$ . Then, if  $r \neq 0$ , it is a prime element if and only if  $\langle r \rangle$  is a prime ideal.

*Proof.*  $\implies$  Let  $r$  be a prime element, and suppose  $ab \in \langle r \rangle$ . Then  $r \mid ab$ , so  $r \mid a$  or  $r \mid b$ , hence  $a \in \langle r \rangle$  or  $b \in \langle r \rangle$  as required.

$\impliedby$  Let  $\langle r \rangle$  be a prime ideal. By definition,  $r$  is not a unit. Say  $r \mid ab$ , so  $ab \in \langle r \rangle$ . Since  $\langle r \rangle$  is prime, either  $a \in \langle r \rangle$  or  $b \in \langle r \rangle$ , so either  $r \mid a$  or  $r \mid b$ . ■

**Proposition 2.35**

Let  $r \in R$  be prime. Then  $r$  is irreducible.

*Proof.* Let  $r \in R$  be prime, and suppose  $r = ab$ . Since  $r \mid r$ ,  $r \mid ab$ , so, since  $r$  is prime,  $r \mid a$  or  $r \mid b$  – wlog suppose  $r \mid a$ . Hence  $a = rc$  with  $c \in R$ , so  $r = ab = rc b$ . Since we're in an integral domain, we can cancel to get  $bc = 1$ , so  $b$  is a unit as required. ■

**Remark.** The converse is false – for example,  $R = \mathbb{Z}[\sqrt{-5}]$ .